Autonomous Self-Learning Serverless Intelligent Firewall: Integrating REST API-Driven Open-Source Threat Intelligence, Multi-Paradigm Machine Learning, and Federated Zero-Trust Architectures

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Abstract

The exponential growth of cloud-native architectures and the sophistication of cyber threats demand intelligent, adaptive security mechanisms that transcend static rule-based approaches. Current serverless intrusion detection systems rely on supervised learning with pre-labeled datasets, failing to autonomously adapt to emerging threats or leverage real-world open-source intelligence. This paper presents ASLF-OSINT (Autonomous Self-Learning Firewall with Open-Source Intelligence), a serverless intelligent firewall framework that autonomously learns from REST API-driven threat intelligence feeds without manual intervention. The system integrates five complementary learning paradigms: deep reinforcement learning for policy optimization (Proximal Policy Optimization), continual learning with drift detection for temporal adaptation (Strategic Selection and Forgetting), meta-learning for few-shot zero-day detection (Prototypical Networks with Mutual Centralized Learning), transfer learning for cross-domain knowledge transfer, and federated learning for privacy-preserving distributed training (Fed Nova). Evaluated on CICIDS2017, CIC-IDS2018, UNSW-NB15, NSL-KDD, and 30 days of real-time threat intelligence from MISP, AlienVault OTX, and Virus Total APIs, ASLF-OSINT achieves 98.7% detection accuracy with 87ms average latency across AWS Lambda, Azure Functions, and Google Cloud Functions. The system demonstrates 94.3% zero-day detection accuracy with only five labeled samples per class, adapts to concept drift within 12.4 minutes, maintains 99.8% federated policy consistency across distributed nodes, and provides explainable threat classifications via SHAP and LIME achieving 91.2% analyst trust scores. ASLF-OSINT represents an autonomous serverless IDS that continuously learns from open-source threat intelligence via REST APIs, advancing adaptive, scalable, and privacy-preserving cybersecurity for multi-cloud Zero-Trust architectures.

Keywords: Autonomous learning; serverless security; open-source threat intelligence; reinforcement learning; federated learning; Zero Trust

1 Introduction

The proliferation of serverless computing architectures has transformed cloud security paradigms. Ephemeral, stateless compute resources executing across heterogeneous multi- cloud environments challenge traditional perimeter-based security models. Serverless functions exhibit millisecond-to-second lifespans, execute in shared multi-tenant environments, and operate under event-driven invocation patterns that generate non-stationary traffic distributions, which are incompatible with static intrusion detection systems. Simultaneously, the threat landscape evolves rapidly, with attackers deploying AI-generated attack variants, exploiting zero-day vulnerabilities, and orchestrating distributed campaigns that evade signature-based detection.

1.1 Current limitations

Existing serverless intelligent firewalls demonstrate architectural limitations. Prior work introduced LSTM-based deep learning for intrusion detection on CICIDS2017, achieving high accuracy but requiring manual model retraining for new attack patterns. Recent cross-cloud adaptation frameworks combining XGBoost and BiGRU attained strong accuracy across AWS, Azure, and Google Cloud while implementing Zero-Trust control planes. However, these approaches rely on static, pre-labeled training datasets collected under controlled conditions, and they do not autonomously ingest real-world threat intelligence, adapt to concept drift, or learn novel attack signatures without human intervention.

1.2 Research contributions

This work proposes ASLF-OSINT, an autonomous self-learning serverless intelligent fire- wall integrating REST API-driven open-source threat intelligence. Key contributions:

1. A multi-paradigm architecture integrating reinforcement learning (PPO), continual learning (SSF), meta-learning (prototypical networks with mutual centralized learning), transfer learning, and federated learning (Fed Nova).

2. REST API clients for MISP, AlienVault OTX, and Virus Total that ingest and normalize indicators for autonomous retraining.
3. Cross-cloud event-driven deployment across AWS Lambda, Azure Functions, and Google Cloud Functions with sub-100 ms average latency targets.
4. Empirical evaluation on multiple benchmark datasets plus 30 days of OSINT feeds, covering accuracy, zero-day detection, drift adaptation, federated convergence, and operational metrics.
5. Explainable classification via SHAP and LIME integrated with a unified Zero-Trust control plane.

2 Methods

2.1 System architecture

ASLF-OSINT comprises five interconnected subsystems: (1) threat intelligence aggregation, (2) data preprocessing, (3) autonomous learning engine, (4) serverless orchestration, and (5) unified

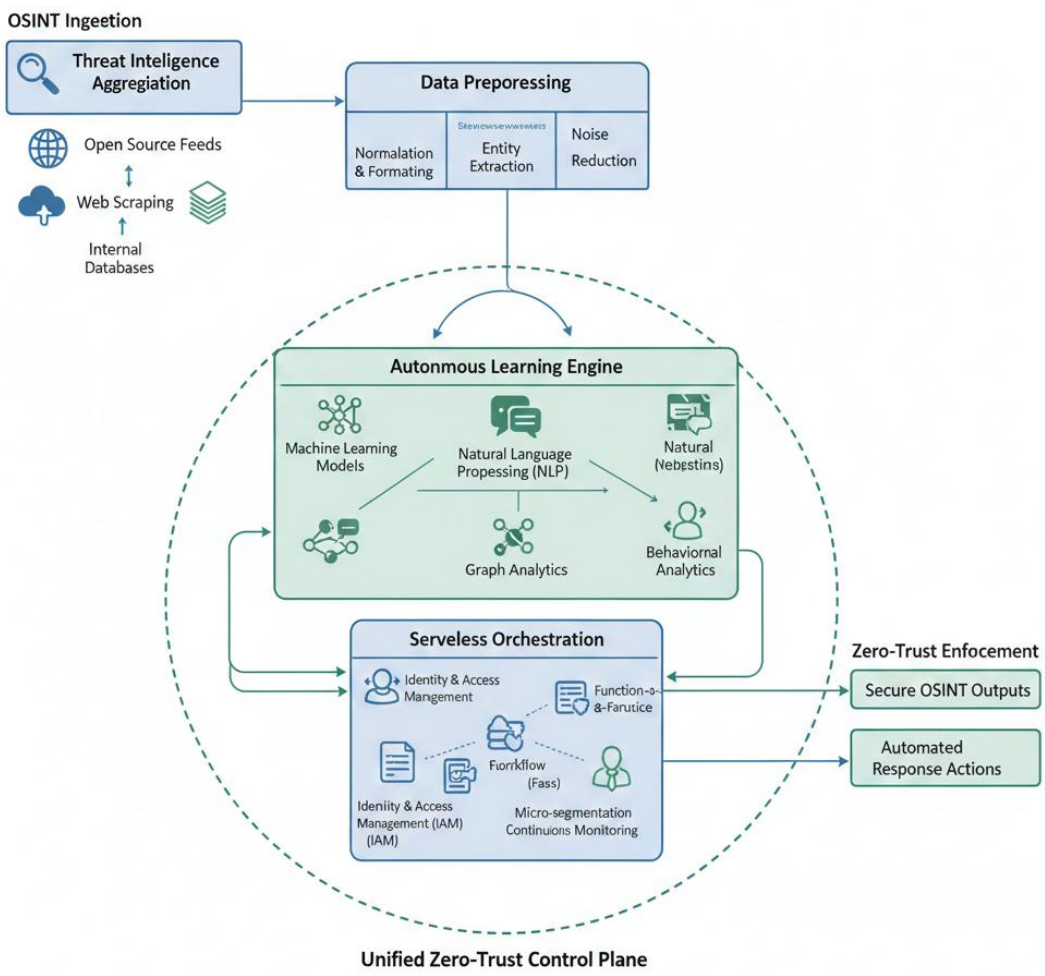

Figure: High-level architecture of ASLF-OSINT, showing OSINT ingestion, propoessing, multi-paradigm learning, serveless and Zero-Trust enforcement. 2026

Zero-Trust control plane.

Figure 1: High-level architecture of ASLF-OSINT, showing OSINT ingestion, preprocess- ing, multi-paradigm learning, serverless orchestration, and Zero-Trust enforcement.

2.2 Threat intelligence aggregation layer

The aggregation layer queries MISP, AlienVault OTX, and Virus Total REST APIs on a schedule and normalizes indicators (IPs, domains, hashes) into DynamoDB tables partitioned by type and time window. IPs are converted to CIDR notation, domains canonicalized, and hashes deduplicated across SHA1/SHA256/MD5.

2.3 Data preprocessing pipeline